



UNIVERSITI
TEKNOLOGI
MALAYSIA
www.utm.my

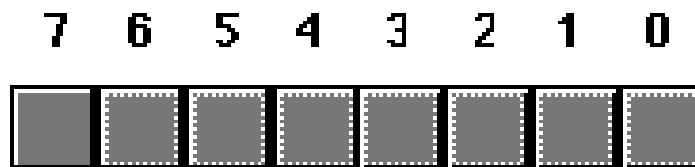
SCSR1013 DIGITAL LOGIC

MODULE 2b: DATA ORGANIZATION (CODES)

FACULTY OF COMPUTING

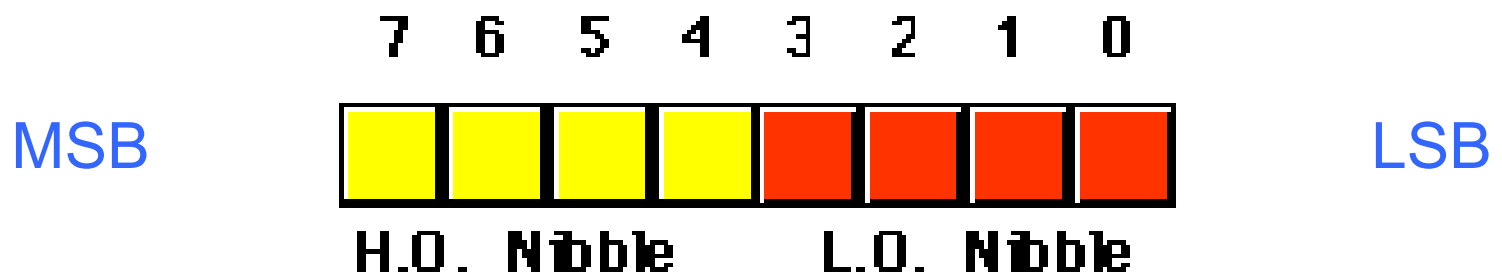
Data Organization

- A value may take an arbitrary number of bits.
- Common collections are single bits
 - smallest "unit" of data on a binary computer is a single [bit](#)
 - groups of four bits called [nibbles](#)
 - groups of eight bits called [bytes](#)
 - groups of 16 bits called [words](#)
- The bits in a byte are normally numbered from zero to seven.



- Bit 0 is the low order bit (rightmost) or least significant bit ([LSB](#)) bit
7 is the high order bit (leftmost) or most significant bit ([MSB](#)) of the byte.

Note **1 byte** also contains exactly **2 nibbles** :



- 4 bits
- **Major uses:**
 - BCD (Binary Coded Decimal)
 - Hexadecimal numbers



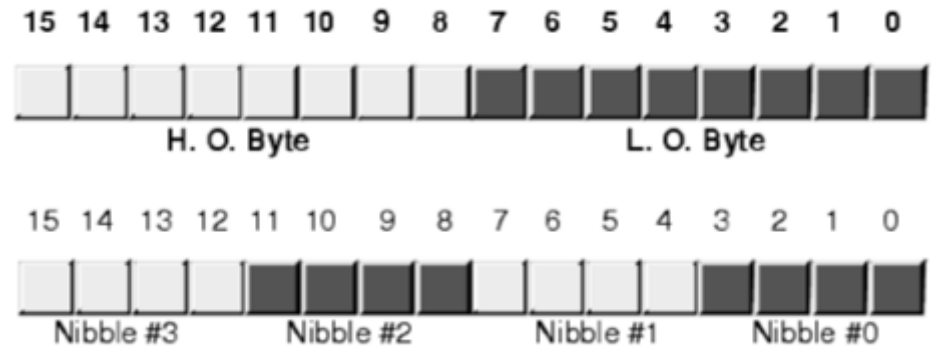
Example: 0111, 1011 and 1111.

7_{16} , B_{16} , F_{16}



- 8 bits
- Total values: $2^8 = 256$
- **Major uses:**
 - Numeric values ($0 \dots 2^8-1 = 0 \dots 255$)
 - Signed numbers: (-128 to +127)

- 16 bits = 2 bytes
- Bit 0 to 15
- **Total values:**
 - $2^{16} = 65,536$



- **Major uses of word:**
 - signed integer (-32,768 ... +32,767)
 - unsigned integer (0 ... $2^{16}-1$) = 0 ... 65,535)
- UNICODE characters

What are codes?

- Code is a representation of information generated by following a certain rules.
- In general, we need code because:
 - Code is unique
 - Codes are easy to process
 - Code is easy to represent
 - Codes enable communication in place where ordinary spoken or written language is difficult or impossible, eg Morse Code
- Due to this, code can simplify the process (such as manipulation and arithmetic operations) of the information in the digital system.

We will learn:

- i. BCD codes
- ii. Gray Codes
- iii. ASCII codes
- iv. Parity codes/bit

Binary Coded Decimal (BCD)

- BCD is a way to express each of the decimal digits with a binary code.
- There are only 10 code groups in the BCD system, one for every digit (0000 – 1001)

Decimal	BCD	Decimal	BCD
0	0000	5	0101
1	0001	6	0110
2	0010	7	0111
3	0011	8	1000
4	0100	9	1001

Invalid codes are 1010, 1011, 1100, 1101, 1110, 1111

Example 1: Convert 3245 to BCD

3 2 4 5
3245 = 0011 0010 0100 0101

Example 2: Convert 7848 to BCD

7 8 4 8
7848 = 0111 1000 0100 1000

Gray Codes

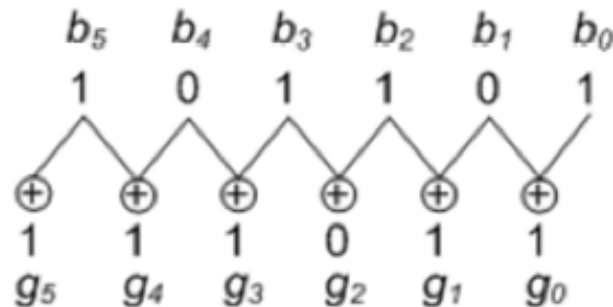
Electromechanical switches



- Designed to prevent **false output** from electromechanical switches.
- Are widely used to facilitate **error correction** in digital communications such as digital terrestrial television and some cable TV systems.
- In modern digital communications, Gray codes play an important role in error correction.
- It is arranged so that every transition from one value to the next value involves only one bit change.
- Sometimes referred to as **reflected** because the first eight values **compare** with those of the last 8 values, but in reverse order.

Conversion of Gray Code $\leftrightarrow$ Binary

To convert **binary $\rightarrow$ Gray code**



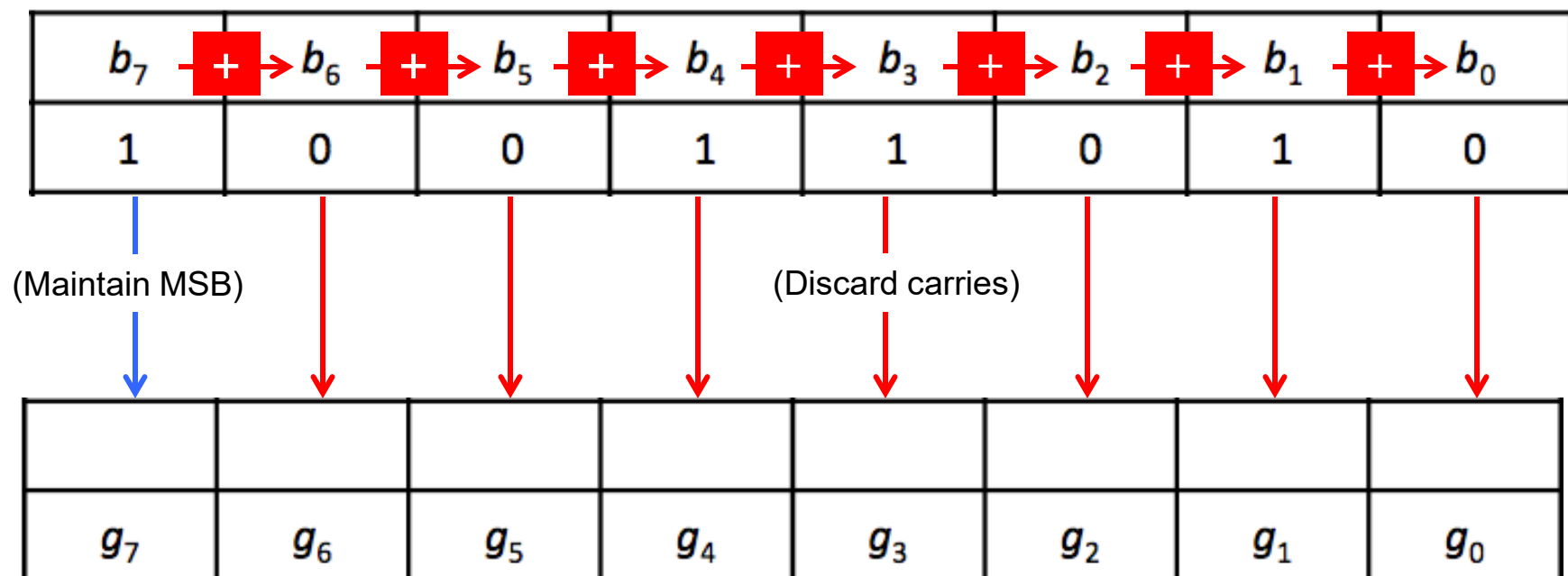
where $b_{i+1}, b_i, b_{i-1}, \dots, b_0$ is the binary number while $g_{i+1}, g_i, g_{i-1}, \dots, g_0$ gray code.

To convert **Gray code $\rightarrow$ binary:**

- $b_i = g_i$ if no. of 1's preceding g_i is even
- $b_i = \overline{g_i}$ if no. of 1's preceding g_i is odd

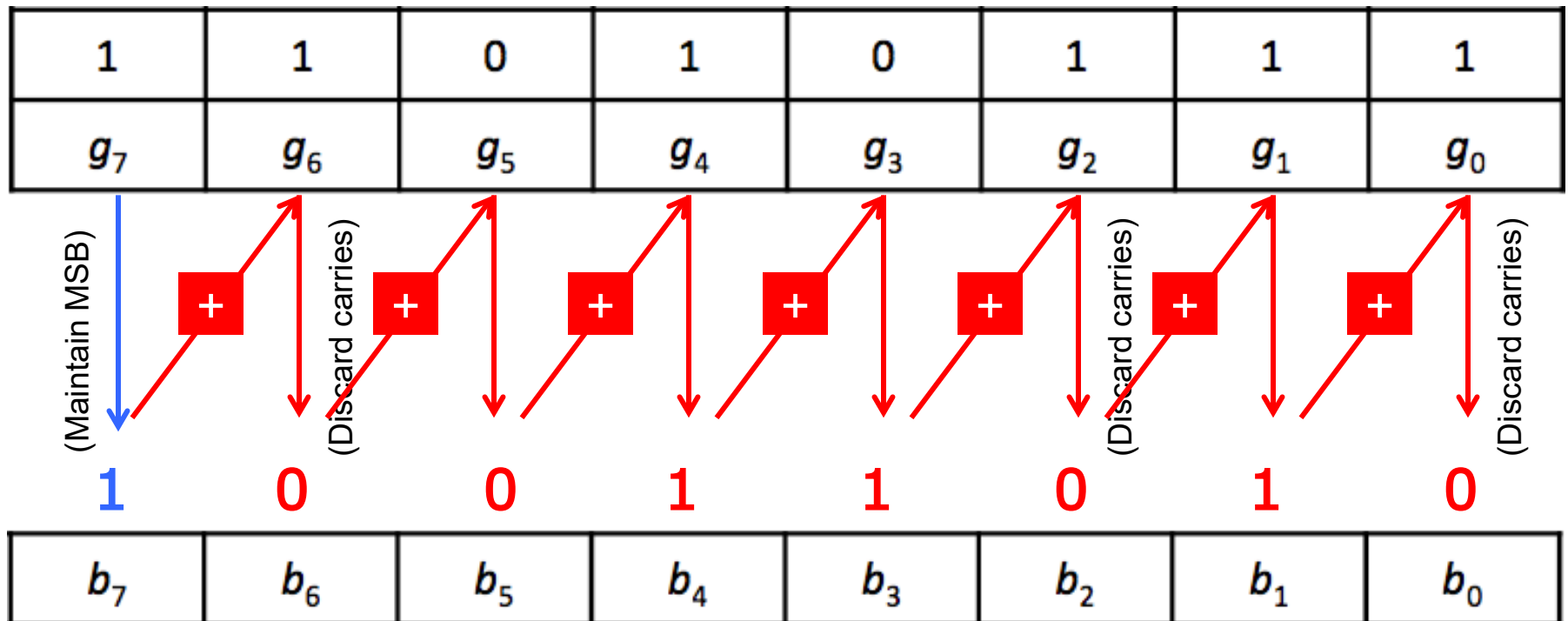
Example:

Convert 100110 to its equivalent gray code value



Example:

Convert the Gray code 11010111 to binary.



Extra

Exercise 2b.1:

Convert 11010100_2 to its equivalent gray code value.

Parity Code

-
- Parity bit used for bit error detection
 - Even parity – total number of 1s even
 - Odd parity – total number of 1s odd
 - Parity bit is append to the code at the leftmost position (MSB) .

Parity bit

0

0

A parity bit is a bit that is added to ensure that the number of bits with value of 1's in a given set of bits is always even or odd. Parity bits are used as the simplest error detecting code.

Extra

Examples:

1 1 0 1 0 0 1 1 1

Even Parity bit

0 1 0 1 0 0 1 1 1

Odd Parity bit

Number 1s	Even Parity	Odd Parity
Even	0	1
Odd	1	0

(Remember these basic rule)

7 bits of data (number of 1s)	8 bits including parity	
	even	odd
0000000 (0)	0 0000000	1 0000000
1010001 (3)	1 1010001	0 1010001
1101001 (4)	0 1101001	1 1101001
1111111 (7)	1 1111111	0 1111111

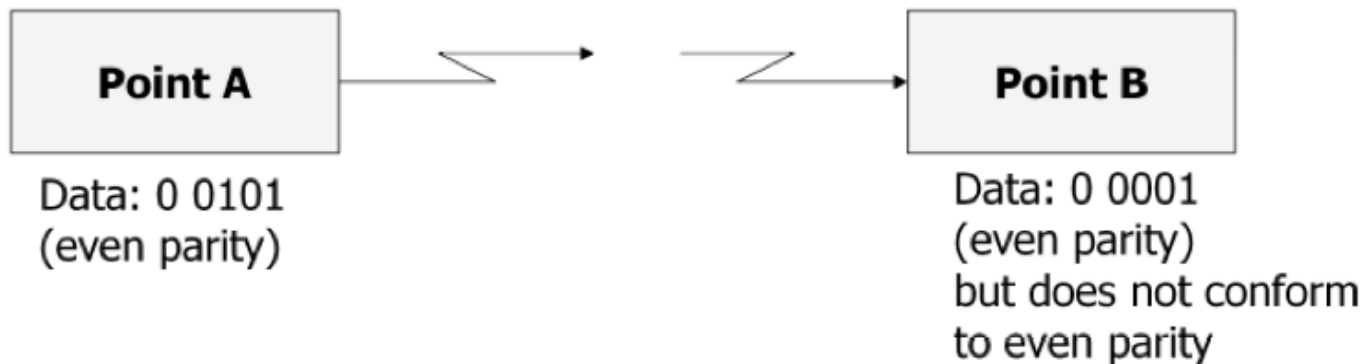
↑ ↑
Parity bit

- Example: Calculate the parity bit for the codes below.

Code	Number of 1s	Even/Odd	Even Parity	Odd Parity
110010	3	Odd	1 110010	0 110010
101110	4	Even	0 101110	1 101110
101000	2	Even	0 101000	1 101000
110111	5	Odd	1 110111	0 110111
111111	6	Even	0 111111	1 111111
100000	1	Odd	1 100000	0 100000

Error Detection by Parity Checking

- Assume that data = 0101
- It uses even parity.
- Therefore the appended parity bit is 0.
- The data with parity bit: 0 0101
- The data is transmitted.
- The data is received as 00001 → odd no. of 1, not even!!



American Standard Code for Information Interchange (ASCII)

- It has 128 characters and symbols represented in 7-bit binary code
- Example :
 - $A = 1000001_2$
 - $a = 1100001_2$
- A parity bit is added so that the total number of bits is 8 $\rightarrow$ a byte.

Decimal	Hex	ASCII	Decimal	Hex	ASCII	Decimal	Hex	ASCII	Decimal	Hex	ASCII
0	00	NUL	32	20	(blank)	64	40	@	96	60	'
1	01	SOH	33	21	!	65	41	A	97	61	a
2	02	STX	34	22	"	66	42	B	98	62	b
3	03	ETX	35	23	#	67	43	C	99	63	c
4	04	EOT	36	24	\$	68	44	D	100	64	d
5	05	ENQ	37	25	%	69	45	E	101	65	e
6	06	ACK	38	26	&	70	46	F	102	66	f
7	07	BEL	39	27	'	71	47	G	103	67	g
8	08	BS	40	28	(	72	48	H	104	68	h
9	09	HT	41	29	)	73	49	I	105	69	i
10	0A	LF	42	2A	*	74	4A	J	106	6A	j
11	0B	VT	43	2B	+	75	4B	K	107	6B	k
12	0C	FF	44	2C	,	76	4C	L	108	6C	l
13	0D	CR	45	2D	-	77	4D	M	109	6D	m
14	0E	SO	46	2E	.	78	4E	N	110	6E	n
15	0F	SI	47	2F	/	79	4F	O	111	6F	o
16	10	DLE	48	30	0	80	50	P	112	70	p
17	11	DC1	49	31	1	81	51	Q	113	71	q
18	12	DC2	50	32	2	82	52	R	114	72	r
19	13	DC3	51	33	3	83	53	S	115	73	s
20	14	DC4	52	34	4	84	54	T	116	74	t
21	15	NAK	53	35	5	85	55	U	117	75	u
22	16	SYN	54	36	6	86	56	V	118	76	v
23	17	ETB	55	37	7	87	57	W	119	77	w
24	18	CAN	56	38	8	88	58	X	120	78	x
25	19	EM	57	39	9	89	59	Y	121	79	y
26	1A	SUB	58	3A	:	90	5A	Z	122	7A	z
27	1B	ESC	59	3B	;	91	5B	[	123	7B	{
28	1C	FS	60	3C	<	92	5C	\	124	7C	
29	1D	GS	61	3D	=	93	5D	]	125	7D	}
30	1E	RS	62	3E	>	94	5E	^	126	7E	~
31	1F	US	63	3F	?	95	5F	_	127	7F	(delete)

Extra

7-bits binary_2
ASCII code

Examples:

$b_7b_6b_5$ $b_4b_3b_2b_1$
110 1101

is represent as
'm'

b_7 b_6 b_5	b_7 → b_6 → b_5 →				b_4 b_3 b_2 b_1				Column → Row ↓			
	0 0 0 0 0 0 1 1 0 1 0 0 0 1 1 0 1 0 0 0 1 0 0 1 1 0 1 0 1 0 1 1 1 1 0 0 1 1 0 1 1 1 1 0 1 1 1 1				0 1 2 3 4 5 6 7 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15				NUL DLE SP 0 @ P ` p SOH DC1 ! 1 A Q a q STX DC2 " 2 B R b r ETX DC3 # 3 C S c s EOT DC4 \$ 4 D T d t ENQ NAK % 5 E U e u ACK SYN & 6 F V f v BEL ETB ' 7 G W g w BS CAN (8 H X h x HT EM) 9 I Y i y LF SUB * : J Z j z VT ESC + ; K [k { FF FC , < L \ l CR GS - = M] m } SO RS . > N ^ n ~ SI US / ? O _ o DEL			

Exercise 2b.2:

Convert the string SCR1013 to its ASCII hexadecimal value.

SCR1013 = 53 43 52 31 30 31 33

By using even parity coding, calculate the parity bit and insert this bit at the MSB position.
Recalculate the ASCII value in its hexadecimal representation.

Exercise 2b.3:

Given a string (character) UTM1435.

- a) Convert the string to its ASCII hexadecimal value.
- b) Calculate the odd parity bit and insert as MSB.
- a) Recalculate the ASCII value in hexadecimal.

Number 1s	Even Parity	Odd Parity
Even	0	1
Odd	1	0

Extra

Character (ASCII)	ASCII (Hex)	Binary	Odd parity bit + Binary	New ASCII (Hex)
U				
T				
M				
1				
4				
3				
5				
h				

Decimal	Hex	ASCII	Decimal	Hex	ASCII	Decimal	Hex	ASCII	Decimal	Hex	ASCII
0	00	NUL	32	20	(blank)	64	40	@	96	60	'
1	01	SOH	33	21	!	65	41	A	97	61	a
2	02	STX	34	22	"	66	42	B	98	62	b
3	03	ETX	35	23	#	67	43	C	99	63	c
4	04	EOT	36	24	\$	68	44	D	100	64	d
5	05	ENQ	37	25	%	69	45	E	101	65	e
6	06	ACK	38	26	&	70	46	F	102	66	f
7	07	BEL	39	27	'	71	47	G	103	67	g
8	08	BS	40	28	(	72	48	H	104	68	h
9	09	HT	41	29	)	73	49	I	105	69	i
10	0A	LF	42	2A	*	74	4A	J	106	6A	j
11	0B	VT	43	2B	+	75	4B	K	107	6B	k
12	0C	FF	44	2C	,	76	4C	L	108	6C	l
13	0D	CR	45	2D	-	77	4D	M	109	6D	m
14	0E	SO	46	2E	.	78	4E	N	110	6E	n
15	0F	SI	47	2F	/	79	4F	O	111	6F	o
16	10	DLE	48	30	0	80	50	P	112	70	p
17	11	DC1	49	31	1	81	51	Q	113	71	q
18	12	DC2	50	32	2	82	52	R	114	72	r
19	13	DC3	51	33	3	83	53	S	115	73	s
20	14	DC4	52	34	4	84	54	T	116	74	t
21	15	NAK	53	35	5	85	55	U	117	75	u
22	16	SYN	54	36	6	86	56	V	118	76	v
23	17	ETB	55	37	7	87	57	W	119	77	w
24	18	CAN	56	38	8	88	58	X	120	78	x
25	19	EM	57	39	9	89	59	Y	121	79	y
26	1A	SUB	58	3A	:	90	5A	Z	122	7A	z
27	1B	ESC	59	3B	;	91	5B	[	123	7B	{
28	1C	FS	60	3C	<	92	5C	\	124	7C	
29	1D	GS	61	3D	=	93	5D	]	125	7D	}
30	1E	RS	62	3E	>	94	5E	^	126	7E	~
31	1F	US	63	3F	?	95	5F	_	127	7F	(delete)